

CHAITANYA PATEL

Milton, ON

(647) 654-4936 | c5patel@uwaterloo.ca | linkedin.com/in/chaitanya-patel-uwaterloo | github.com/chaitanyaCE

Education

University of Waterloo

September 2025 – May 2030 (expected graduation)

Honours Computer Engineering, Co-op

Waterloo, ON

- **Relevant Coursework:** Programming & Software Development (Object-Oriented C++), Engineering Design, Discrete Mathematics

Skill Set

Programming Languages: Python 3.x, C/C++, SQL, SQLite, JavaScript (ES6+), TypeScript, React, Node.js, Puppeteer Library (JavaScript)

Data & Systems Engineering: Data Structures & Algorithms, REST API Concepts, JSON Data Parsing, Object-Oriented Programming (OOP), Exception Handling

Governance, Cloud & Security: AWS Cloud Architecture, Microsoft Azure Cloud Concepts, Cloud Security Architectures, Governance, Risk Management, and Compliance (GRC), Infrastructure Management

Methodologies & Tools: Git & GitHub, Linux Environments, SQLite, Figma, Bash/Shell Scripting Awareness, Software Development Life Cycle (SDLC), Agile (Scrum)

Relevant Experience

Web & UX Intern | Waterloo Experience Accelerate Program

January 2026 – April 2026

Grassroots Response To The Ukrainian Crisis

Waterloo, Canada

- Diagnosed and resolved critical system bugs by executing **manual QA test scenarios and redesigning UI/UX elements** across multiple deployment environments to ensure code quality and platform reliability for over 1,000 monthly active users.
- Accelerated the agile bug resolution process by authoring detailed **technical defect reports and internal documentation** to support the core engineering team's long-term maintainability efforts.
- Translated user requirements into actionable technical implementations by collaborating actively in **daily Agile sprint planning ceremonies and stand-ups** to improve overarching system performance.

Projects

Expense Tracker | Vanilla JavaScript, Core State Routing, UI Engineering

- Built a custom financial calculation engine to master core **logic routing, algorithmic efficiency, and manual data manipulation**, strictly avoiding high-level frameworks to understand underlying system behaviors.
- Designed a **dynamic profit percentage calculator** that processes real-time transaction data, implementing algorithmic checks to automatically update system states based on mathematical thresholds.
- Wrote custom logic to correct floating-point calculation errors, applying a rigorous performance analysis mindset to manually **test over 15 distinct edge cases** and guarantee zero data corruption.
- Handled data persistence securely, writing robust *try...catch* blocks to **parse complex JSON objects and prevent system crashes** from malformed memory strings.
- Utilized an **AI-driven workflow via GitHub Copilot** to iterate on logic efficiency, tracking all codebase updates securely using Git to maintain a strict version control history.

TaskFlow SPA | DOM Manipulation, State Management, Git/GitHub Version Control

- Programmed a responsive application to instantly **process, categorize, and render dynamic data sequences** using core algorithmic array sorting methods (*.map*, *.filter*).
- Developed **custom conditional logic** to optimize interface rendering, automatically updating the view state to handle empty data returns gracefully without breaking system execution.
- Designed robust **client-side data management** to handle dynamic sequences containing over 40 active items, ensuring seamless transitions and efficient memory handling.
- Maintained a clean audit trail in a public GitHub repository, organizing codebase updates into focused commits to strictly isolate algorithmic changes.

Certifications

Governance & Cloud Infrastructure: Microsoft Security Essentials Professional; Microsoft Certified: Azure Fundamentals Cloud Concepts (AZ-900); AWS Certified Cloud Practitioner (In Progress)

Core Engineering & Data: Data Structures & Algorithms; Python Essential Training; SQL Essential Training